

characteristics relative to some traditional aspects of enterprise IT. But at the same time, most organisations are still apprehensive about moving all their applications onto public cloud providers. Whatever the reasons for that may be, there is nevertheless also considerable interest in the idea of building hybrid clouds spanning both on-premise and off-premise resources to deliver the best of both worlds: public cloud economics and agility optimised for enterprise needs such as audit, risk management, and strong policy management.

Careful evaluation needs to be part of the decision-making process when adopting a private, hybrid, or publicly hosted cloud model. Large enterprises and government IT departments have long utilised costly, proprietary systems as their core IT systems. Often, an important step on the journey to cloud computing will be to standardise legacy systems, as appropriate, onto modern, standards-based hardware and software. Without taking an inventory of and developing a plan for their existing IT infrastructure and applications, organisations put themselves at a disadvantage of not being able to fully realise the financial and technical benefits of cloud computing. For example, adopting an open architected and standards-based operating system such as Linux, which is built on the principles of community-powered innovation, provides many benefits. These enterprise-class open source offerings can provide a solid foundation for next-generation IT infrastructure and offer world-class support so organisations can achieve better business performance and greater cost efficiencies.

An open, hybrid cloud management approach allows organisations to evolve to pursue this sort of standardisation at their own pace. It offers an alternative to replacing all their existing applications and infrastructure, and starting anew—a tempting approach to simplification but one that's not practical for most. It also offers a better way than simply extending an existing set of proprietary technologies and products, such as a virtualisation platform, onto the cloud. Doing so can simply transfer an organisation from one form of vendor lock-in to another, trading proprietary virtualisation for the proprietary cloud.

Openness provides a path in this evolutionary approach that puts organisations in control of their technology roadmap and the future of their IT resources. We see open source as part of the equation.

Open source vendors contribute source code to projects and let all users contribute to make it better. The widespread support of KVM (Kernel-based Virtual Machine) and the Open Virtualization Alliance (OVA) by organisations that typically compete, including IBM, HP, Dell, AMD, Intel and Fujitsu, is a testament to their interest in an open approach for hypervisors. The oVirt project adds open source virtualisation management

Hybrid cloud management, which allows organisations to build a cloud out of multiple virtualisation platforms as well as public clouds, can fundamentally benefit from open source and other aspects of openness such as open standards and open application programming interfaces (APIs). A good example is the Deltacloud project, which is governed by the Apache Software Foundation. It offers a transparent, 'no lock-in' approach to cloud-building by providing a set of APIs that can talk to a heterogeneous set of 'cloud providers', including virtualisation stacks and public clouds. Any decision to move to the cloud needs to take into account the ease of management in a cloud and across different clouds, as well as the ability to change cloud providers if and when an organisation chooses to do so.

Cloud computing has certainly reached a level of market maturity and we are well on our way to a stage where it will become the accepted norm. In the meantime, for many, the cloud is still a thrilling new technology alternative that holds immense promise but should be approached with the same deliberation and thought as with any other strategic IT product. And make no mistake about it—cloud computing is, or should be, strategic for your organisation. 

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